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Scrum Master: Mineh Jami

Developers: Sonbol Almas, Sewara Rashidi

Project: Clinic Appointment Management System

Sprint: Sprint 2 (Oct 16-21, 2026)

Sprint Goal:

Deliver the doctor search and filtering page, including the doctor information and availability section. Allow an authenticated patient to be able to view current doctor appointment dates and time slots, and get the system ready for integration of the appointment booking feature.

User Stories:

ID	User Story	Story Notes	Status
US-05	The system shall allow patients to search for doctors.	Patients can search for doctors by name or specialty	Done
US-06	The system shall allow patients to view doctor information and availability.	Patients can view a doctor's profile, including professional details, specialty, and availability.	Done
US-07	The system shall allow patients to view available appointment dates and time slots.	Patients can view available appointment dates and time slots to choose a suitable appointment time.	Done
US-15	As a doctor, I want to manage my availability so that patients can book appointments during my available times	Doctors can add or remove dates and time slots to determine availability.	Done

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Acceptance Criteria:

US-05: Doctor Search

1. Patient can enter search criteria.
2. System displays matching doctors.
3. Patient can select a doctor to view their information.

US-06: Doctor Information and Availability

1. Patient can view the doctor's professional information.
2. Patient can view the doctor's available dates and times.
3. The displayed availability reflects the doctor's current availability

US-07: View Available Appointment Dates and Time Slots

1. System displays available appointment slots.
2. Patient can select an available slot.
3. System confirms the appointment after successful booking.

US-15: Doctor Availability Management

1. Doctor can set available dates and times.
2. Doctor can update their availability.
3. Updated availability is reflected when patients view appointment slots.

Sprint Tasks

Task ID	Task description	User Story	Status
T2.1	Design the doctor search interface with search fields and results layout.	US-05	Done
T2.2	Implement backend search functionality by doctor name and specialty.	US-05	Done
T2.3	Display matching doctors and handle searches with no results.	US-05	Done
T2.4	Create the doctor information and availability details page.	US-06	Done
T2.5	Retrieve and display doctor information from the database.	US-06	Done

T2.6	Retrieve and display available appointment dates and time slots.	US-06	Done
T2.7	Implement the interface for viewing available appointment dates and time slots.	US-07	Done
T2.8	Implement date selection and display available time slots for the selected date.	US-07	Done
T2.9	Ensure unavailable or already-booked slots are not displayed as available.	US-07	Done
T2.10	Implement functionality for doctors to manage their available days and time slots.	US-15	Done
T2.11	Save and retrieve doctor availability from the database.	US-15	Done
T2.12	Prevent doctors from creating overlapping availability time slots.	US-15	Done
T2.13	Test doctor search, profile viewing, availability display, profile updates, and access permissions.	US-6 to US-15	Done

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Definition of Done:

A user story can be considered Done when:

- All its acceptance criteria are satisfied.
- The implementation is integrated into the working product.
- Relevant tests pass.
- No unresolved critical defects prevent the functionality from being used.
- Security requirements relevant to the functionality are satisfied.
- The completed work meets the team's agreed quality standards.

Sprint Metrics:

Metric	Target	Actual	Status
User stories completed	4 selected user stories	4 completed stories	Done
Bugs Found	Track all identified bugs	All bugs were identified	Done
Bugs Fixed	Fix relevant defects identified during the Sprint	All identified bugs were fixed	Done
Required Tests	All required tests should pass	The 18 tests performed passed	Done
Sprint Goal Met	Achieve the Sprint Goal	Sprint goal was met	Done

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Increment/Sprint Outcome:

Working functionality delivered:
The Sprint 2 increment focused on enabling patients to search for doctors by name or specialty, view doctors' professional information and appointment availability, and allowing doctors to manage their available days and time slots. The planned functionality also included saving availability changes to the database, validating inputs, and ensuring patients could view updated availability.

Overall outcome:
Sprint 2 aimed to improve the Clinic Appointment Management System by connecting patients with suitable doctors and providing a foundation for appointment scheduling. The increment was intended to build on Sprint 1's authentication and role-based access functionality, preparing the system for future features such as appointment booking, double-booking prevention, and appointment confirmation. Final completion should be confirmed against the Sprint Backlog and the team's Definition of Done.

Testing and quality summary:

Sprint 2 Day 4 regression testing was conducted by Sewara Rashidi and Sonbol Almas using PHP, MySQL, XAMPP, and a web browser. A total of 18 manual integration and regression tests were executed, and all 18 passed with zero failures. Testing covered doctor search, professional profile and availability display, appointment date selection and filtering, appointment booking, database persistence, and protection of existing appointments when modifying or deleting doctor

availability. Database verification confirmed that booked appointments retained their confirmed status and that booked time slots remained unavailable after refreshing the page. No failures were recorded in the tested scenarios. However, automated regression testing, simultaneous booking requests, overlapping availability periods, appointment cancellation and rebooking, and cross-browser/device testing remain unverified. Final acceptance of the Sprint 2 user stories is subject to the team's Definition of Done and Sprint Review.

Stakeholder Feedback:

None was recorded.

Issues and Blockers:

Issue: My MySQL in XAMPP stopped working several times. At first the problem was caused by corrupted Aria log files which was fixes in the previous sprint by the developer, but the error happened again on different days.

Resolution: Another error was found in the mysql.db system table, after the developer repaired the damaged table using Command Prompt, MySQL started working again.

Issue: Another developer also had problems starting MySQL in XAMPP, however, her problem was different. Port 3306 was already being used by another MySQL process, which had happened in the previous sprint as well, and so XAMPP could not start MySQL.

Resolution: Once MySQL was restarted everything started working again.

Product Backlog Updates:

No updates or refinements to the original document were made, but some clarification and confirmations were made by the product owner during a meeting together with one of the team developers. Namely, what doctor information should be seen by the patients, what category they should search the doctor by, and how doctor availability should work. There were also clarifications regarding appointment durations, editing and deleting availability, and staff approval requirement for confirming booking.

Sprint Retrospective — Improvement Action:

Improvement identified:

- Complete sequence and UML diagrams earlier in the Sprint rather than leaving them until the end.
- Identify database changes and backend–frontend dependencies early to reduce integration issues.
- Review planned interfaces together before implementation to ensure both developers agree on the design and understand the information required from the backend.
- Conduct usability testing from the perspective of someone unfamiliar with the system to identify confusing workflows and overlooked issues.

Action for the next Sprint:

- Prepare and review sequence and UML diagrams during Sprint planning or at the beginning of the Sprint.
- Discuss database structure, backend requirements, and frontend dependencies before starting implementation.
- Review and approve interface designs collaboratively before dividing development tasks.
- Integrate frontend and backend components incrementally throughout the Sprint instead of waiting until the end.
- Ask someone who was not involved in development to test important user actions and provide feedback before the Sprint Review.